

The year 2023 marks a turning point for me. Generative AI has changed my life. It's just like how the X-ray was invented in 1895, which opened a door to many discoveries later in the 20th century. Now, people who aren't smart can also do great work. where a discovery mindset and motivation for innovation are prioritized over technical skills more than ever before. I have been following the development in the GenAI field very closely. My work will be AI-integrated.

Work

- **Instructional Student Assistant / Teaching Associate - San Diego State University 01/2025 - 12/2025**
 - Covered physics lab for first year students.
- **Research Assistant for Precision Swine Production - University of Missouri 07/2024 - 12/2024**
 - Developed solutions to predict estrous time in swine: configured MCU, and LIDAR/IR cameras with codes to become an inspection robot; trained in models for imaging segmentation and transformer neural network. Filled an NSF fellowship proposal; a two-page extended abstract proposed that the ground truth uncertainty issue might be addressed by using a microneedle patch for continuous estradiol monitoring
- **Bioinformatics Research Assistant - Florida Atlantic University/Texas Tech University, 03/2023 - 06/2023, 04/2024 - 06/2024**
 - Performed mutation and transposable element between samples from whole exome/genome data, using GATK/BWA/Samtools/Bcftools/Pysam/SnpEff/customized python code, short-term contractor working with the same team.
- **Research Data Assistant - IU School of Medicine, 06/2021 - 07/2022**
 - Analyzed over 50 various RNA-seq projects at the sequencing core, including: processing raw RNA sequencing data, mapping, sample outlier detection, batch effect removal, various DGE analysis, pathway analysis, splicing analysis, and troubleshooting
 - Generated html report with the newest visualization tools/plots for the users
 - Processed raw WES data, SNP, INDEL analysis
 - Developed a GUI database
 - Provided consultation to researchers regarding the analysis results
 - Mentored undergraduate and master's students

Education

- **Arizona State University, 01/2025 – present, M.S. Electrical Engineering, online**
Took Digital Circuit, FPGA, VLSI design, Digital Signal Processing (DSP) courses
 - **San Diego State University, 01/2025 – 12/2025, M.S. Medical Physics, onsite**
CAMPEP program, clinical practicum for compliance QC in radiography, fluoroscope, CT, SPECT, ultrasound; machine QA, patient specific QA; TG66/TG51/TG61/TG148/TG306. Research in blockchain data systems for agriculture and medical imaging applications and MR imaging reconstruction through DL models. Selected for two fellowships in research. Took upper-level physics courses equivalent to physics minor prior enrolling.
 - **Purdue University, 08/2023-05/2024, non-degree student taking medical physics courses,**
Took courses about how imaging generated through X-ray, CT, PET, MR, or ultrasound; conducted MR research that isolated non-water proton signals, led in a poster presentation at the school conference; developed TG51 online calculator for radiation dosage calculation led to oral presenting in *regional* conference
- 2021
- **Indiana University – Purdue University Indianapolis (IUPUI), B.S. in Biomedical Informatics,**
About 150 credits in biotech, statistics, computer science, bioinformatics. Internship to identify

potential pathways and markers related to mitochondria in RGC cells from single-cell sequencing data.

Projects

- **Genomics Bioinformatics (2020-2023):** Whole genome SNP/INDEL calling, transposable-element mapping, large-scale RNA/miRNA/miRNA/lncRNA pipelines, multi-generation family inheritance mutation calling, possible biological meaning with literature search; single cell functional check https://ax3301.github.io/RGC/rgc_2021.html; dashboard development https://ax3301.github.io/PDAC_2022
- **Web Scraping + Functional call: Toolkit for new medical physics students** https://ax3301.github.io/Medical_Physics_Toolkit_23/
- **NSF GRFP Proposal 2023: Large language models for advancements in biomedical research** https://ax3301.github.io/RGC/NSF_2023.pdf
- **NSF GRFP Proposal 2024: Precision livestock farming using engineering strategies in swine Barns** https://ax3301.github.io/RGC/NSF_2024.pdf
- **Water signal suppression in NMR imaging** https://github.com/ax3301/RGC/blob/main/NMR_2024.ipynb
- **Swine Vision: Imaging segmentation and time series transformer neural network in swine** https://github.com/ax3301/Pig_Vision_2024
- **NSF New Grants Tracker and AI Agent** <https://nsf-grant-tracker-2025.sequencingmaster.com/>
<https://nsf-grant-tracker-2025.sequencingmaster.com/agent>
- **California State University Agriculture Research Institution Student Fellowship Proposal 2025: Blockchain-enabled agricultural data system with AI for swine and soybean production (Selected for 10k)** https://ax3301.github.io/RGC/F_2025.pdf
- **Developing Blockchain Workflow for Quality Control in Medical Physics_2025** https://docs.google.com/presentation/d/1UhgE93MdXfVbx7gheiY_hSyWWwMQ-oUh/edit?usp=sharing&ouid=111482565369766842097&rtpof=true&sd=true <https://imaging-mp-record.pages.dev/>

Publication, Poster, Oral Presentation

- Chang H. (2021). A comprehensive list of undergraduate bioinformatics programs in the United States. <https://doi.org/10.31219/osf.io/9g5wu>
- QiuHong He, et al. In Vivo pi-SSeIMQC to Recover 100% Biomarker Signals From Both MQ Coherence Transfer Pathways with Applications in Tumor Lactate Imaging. Academic Radiology. <https://doi.org/10.1016/j.acra.2025.11.042>
- Chang H. (Apr 2024). The purpose of developing an online TG51 calculator for quick beam measurement: To provide a convenient and accessible tool for trainees. Oral presentation at: American Association of Physicists in Medicine ORVC 2024 Spring Meeting. https://ax3301.github.io/RGC/ORVC_2024.pdf

Technical Skills

Languages & Tools: Python, R, MATLAB, VHDL, SystemVerilog, HTM, Bash

Hardware & EDA: Microcontrollers, LiDAR/IR sensors, Virtuoso, Vivado

AI & ML: PyTorch, Transformers, OpenCV, Prompt Engineering

Bioinformatics: GATK, BWA, SAMtools, BCFtools, SnpEff, Pysam, Seurat, DESeq2

Data & Visualization: Pandas, Plotly, Dash, Matplotlib, ggplot2